AI based Intelligent Framework for 3DCP Layer Deformation Monitoring and Shrinkage Prediction

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***Abstract*—3D Concrete Printing (3DCP) is an emerging additive manufacturing technology that enables automated, layer-by-layer construction of complex concrete structures, reducing labor, material waste, and construction time. Despite its potential, 3DCP faces critical challenges related to layer stability, process control, and long-term durability due to shrinkage and crack formation. This study focuses on three key aspects: (1) determining the maximum number of layers that can be printed before lower layers deform, (2) developing a real-time machine learning-based system to monitor local deformations and adjust printing parameters dynamically, and (3) predicting shrinkage and crack initiation using process, material, and environmental data. By integrating experimental observations with predictive modeling and adaptive control, the proposed framework aims to enhance the reliability, structural performance, and efficiency of 3DCP, paving the way for its wider adoption in modern construction practices.**

***Index Terms*—3D Concrete Printing, Additive Manufacturing, Layer Stability, Machine Learning, Shrinkage Prediction, Crack Formation**

I. Introduction

Concrete is the most widely used construction material in the world, forming the backbone of modern civilization. From buildings and bridges to roads and dams, it is an indispensable component of urban infrastructure and global economic development. Despite its ubiquity and versatility, conventional concrete construction methods have remained largely unchanged for decades. These traditional processes are often time-consuming, labor-intensive, and susceptible to human error. Moreover, they rely heavily on formwork, which not only increases material consumption and costs but also restricts the architectural freedom of designers and engineers.

In recent years, the construction industry has been undergoing a digital transformation, driven by the need for greater efficiency, sustainability, and design innovation. Among the most promising developments is 3D Concrete Printing (3DCP), an advanced additive manufacturing process that fabricates concrete structures layer by layer through the controlled extrusion of a specially formulated cementitious material. By eliminating the need for formwork and manual placement, 3DCP significantly reduces labor costs, minimizes material waste, and accelerates construction timelines. Furthermore, it enables the creation of complex geometries, organic shapes, and customized architectural forms that are often impractical or impossible to achieve through conventional casting methods.

Research and industrial applications have demonstrated that 3DCP can potentially reduce overall construction costs by 50%–60%, while simultaneously enhancing sustainability by optimizing material use and reducing the carbon footprint of construction projects. However, realizing these benefits on a large scale requires a thorough understanding of the interplay between material science, mechanical design, and process control. The printed structure's quality depends not only on the concrete mix design but also on the precise calibration of printer parameters such as nozzle diameter, extrusion pressure, print speed, and layer height.

During a recent visit to the 3DCP Laboratory at the National Institute of Technology Karnataka (NITK), Surathkal, we observed the complete working mechanism of a large-scale 3D concrete printer. The visit provided valuable insights into the key factors influencing the printing process, including the rheological behavior of the printable mix, the bonding strength between consecutive layers, and the influence of environmental conditions such as temperature and humidity. It was evident that achieving a successful print is not solely dependent on advanced hardware but equally on the optimization of material properties and real-time control of process parameters.

One of the major challenges in 3DCP lies in developing an ideal mix design that meets multiple performance requirements simultaneously. The fresh concrete must possess adequate pumpability and extrudability to ensure smooth and continuous flow through the nozzle without clogging. At the same time, it must maintain sufficient buildability and yield stress to support subsequent layers without excessive deformation or collapse. Achieving this delicate balance requires fine-tuning of mix proportions, admixtures, and additives that influence the setting time, thixotropy, and interlayer adhesion.

Furthermore, printer parameters such as extrusion pressure, nozzle travel speed, and interlayer time intervals play a critical role in determining the dimensional accuracy and mechanical integrity of the printed structure. Small deviations in these parameters can lead to layer misalignment, poor bonding, or structural instability. To overcome such issues, modern 3DCP systems are increasingly integrating machine learning and real-time monitoring techniques to adaptively control the

printing process and predict material behavior under varying conditions.

In this context, the present study focuses on three critical aspects of 3D Concrete Printing: (1) ensuring layer stability during the printing process to prevent collapse or misalignment, (2) developing a real-time adaptive control system using machine learning for process optimization, and (3) predicting shrinkage and crack formation in printed elements to improve long-term durability. Addressing these challenges will contribute toward the advancement of reliable, automated, and sustainable 3D concrete construction technologies capable of reshaping the future of the construction industry.



Fig. 1. Example of a 3D Concrete Printing setup.

II. Background

The conventional construction industry has long faced persistent challenges, including low labor efficiency, high accident rates, a diminishing skilled workforce, and rising costs associated with time-consuming processes [1]. A significant portion of these costs and delays is attributable to the creation and removal of formwork, a necessary but wasteful and labor-intensive step in traditional concrete casting [2]. In response to these issues, 3D Concrete Printing (3DCP) has emerged as a paradigm-shifting technology, leveraging digital fabrication to automate construction processes [3]. As a form of additive manufacturing, 3DCP builds structures by depositing a specially formulated cementitious material layer-by-layer directly from a digital model. This method's primary advantage is the elimination of formwork, which offers the potential for accelerated construction timelines, reduced labor costs, minimized material waste, and unprecedented geometric freedom in architectural design [2][4].

The foundational work in this field was pioneered through distinct, yet complementary, research streams that defined the technology's primary development paths. The seminal work by Behrokh Khoshnevis at the University of Southern California introduced *Contour Crafting*, a large-scale, on-site extrusion-based method designed for the rapid construction of entire buildings [1]. This approach laid the theoretical and technical

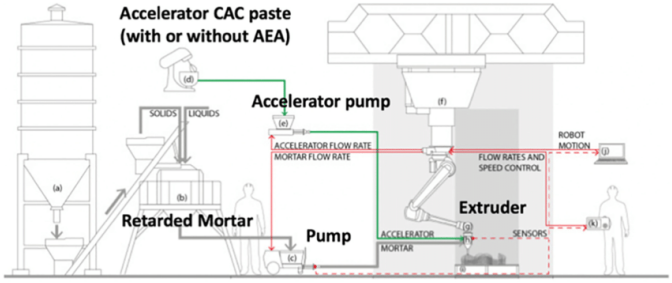


Fig. 2. 3D Concrete Printing Workflow

groundwork for automated construction. A key innovation of Contour Crafting was the integration of robotic trowels adjacent to the extrusion nozzle, which smooth the surfaces of the deposited layers to create a high-quality, paint-ready finish directly within the printing process, while also allowing for the embedding of conduits for electrical and plumbing services [4].

III. Literature Review

The application of Machine Learning (ML) to address the inherent variabilities and challenges in 3D Concrete Printing (3DCP) is a burgeoning field of research. Predictive modeling, in particular, has shown significant promise in optimizing mix designs and forecasting the mechanical performance of printed structures, thereby reducing the need for extensive and time-consuming experimental trials [6].

Several studies have focused on predicting the final hardened properties of 3DCP. For instance, Ali et al. [1] developed various ML models, including Support Vector Machine (SVM) and XGBoost Regression, to predict the flexural and tensile strength of 3D printed concrete [7]. Their work utilized a comprehensive dataset of mix design parameters such as water-to-binder ratio, cement content, and fiber properties as inputs. The study successfully demonstrated that the SVM model could predict the mechanical properties with high accuracy, establishing a strong precedent for using ML to navigate the complex, multi-variable problem of mix design optimization.

Building on this approach, Zhang et al. [2] also employed a suite of ML algorithms, including random forests and artificial neural networks, to predict the mechanical properties of 3D printed concrete, achieving high correlation coefficients [8]. Their research further underscores the capability of ML to model the complex non-linear relationships between material composition and structural performance. More recent work has even extended this approach to predict critical process-related properties, such as the interlayer bonding strength (IBS), using advanced ensemble techniques like stacking models to improve predictive accuracy over single-model approaches [9].

While these studies provide a robust framework for predicting final mechanical strength based on initial mix proportions, a significant gap remains in the real-time monitoring and prediction of dynamic behaviors during the printing process itself. The critical challenges of early-age layer deformation under loading from subsequent layers, and the long-term

effects of shrinkage, are highly dependent not only on the mix design but also on dynamic process parameters (e.g., print speed) and ambient environmental conditions [10]. The existing literature has not fully addressed the development of adaptive ML systems that can predict these transient phenomena in real-time to actively control the printing process. This study aims to bridge that gap by developing intelligent models focused specifically on predicting layer deformation and shrinkage, integrating process and environmental data to enable a dynamic, adaptive control system for 3DCP.

A. Problem Statement

To develop AI-based predictive and control models that can analyze material behavior, monitor local deformations, to enable dynamical adjustment of printing parameters such as extrusion rate and print speed to mitigate shrinkage and improve interlayer bonding performance in 3DCP structures.

B. Objectives

1) Detailed study of parameters and features required for predicting shrinkage in 3DCP structures
2) Data collection and design of preprocessing pipeline for collected data.
3) Developing AI models to analyse and evaluate local deformations during printing and dynamically adjust printing speed or extrusion rate to maintain structural stability.
4) Predicting shrinkage behavior and the onset of crack formation in printed structures to ensure long-term durability and interlayer bonding integrity.

IV. Proposed Methodology

Fig. 3 depicts the overall process of the proposed workflow for developing a machine learning-based adaptive control model for 3D concrete printing (3DCP). The methodology comprises four main stages — Dataset Collection, Feature Engineering, Predicted Parameters, and Prediction Model Design — which collectively represent the data-driven control pipeline.

A. Dataset Collection

The dataset for this study will primarily be obtained from the Civil Engineering Department, NITK (3DCP Laboratory), which maintains experimental records of 3D concrete printing (3DCP) processes, material mix designs, and mechanical test data. To enhance data diversity and model generalization, supplementary data will also be incorporated from open-source repositories and published studies [1]–[3]. The dataset will comprise both *input features (independent variables)* and *output features (dependent variables)* relevant to layer stability, adaptive printing control, and shrinkage prediction.

B. Feature Engineering

The input features encompass material composition, printing process parameters, and rheological properties that influence the mechanical and structural performance of printed layers.

- Material Composition:

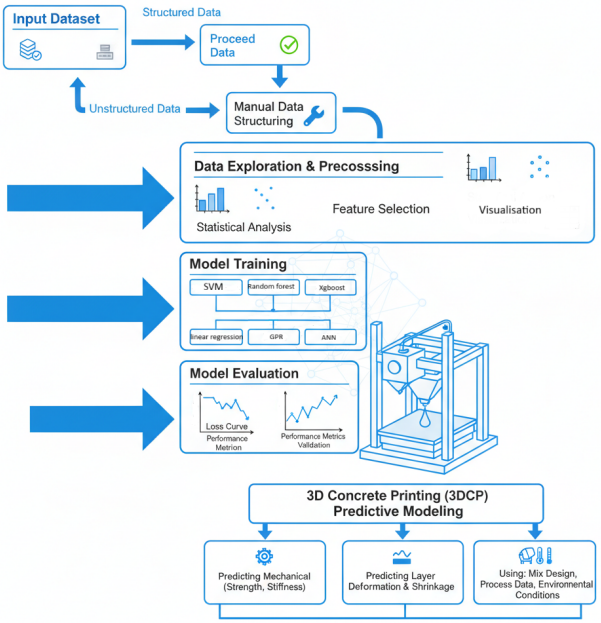


Fig. 3. Proposed Workflow

  - Cement content (kg/m^3) primary binder influencing hydration rate and compressive strength [1].
  - Water content (kg/m^3) and water–binder ratio (w/b) control workability and layer adhesion [2].
  - Silica fume, fly ash, and slag (kg/m^3) supplementary cementitious materials that enhance microstructure and reduce shrinkage [3].
  - Fine and coarse aggregates (kg/m^3) and particle size (mm) influence rheology, buildability, and interlayer support.
  - Fiber content (kg/m^3), diameter (mm), and tensile strength (MPa) enhance crack resistance and flexural stability [1].
  - Viscosity-modifying agent and superplasticizer (kg/m^3) control extrudability, shape retention, and layer cohesion.
- Process Parameters:
  - Print speed (mm/s) affects extrusion uniformity and deformation rate under load [1].
  - Nozzle area (mm^2) and layer height (mm) determine flow rate and bonding surface area.
  - Interlayer time interval (s) influences adhesion strength and shrinkage cracks at layer interfaces.
  - Extrusion pressure (Pa) controls filament consistency and layer thickness stability [2].
  - Ambient temperature and relative humidity affect setting time and drying shrinkage [3].
- Rheological and Mechanical Properties:
  - Yield stress (Pa) and plastic viscosity (Pa·s) govern extrudability and build-up rate of printed layers.

  - Buildability index and deformation rate indicators of print stability and self-weight resistance.

C. Predicted Parameters

The dependent variables, representing the measurable responses of the printed concrete, will be used to evaluate model performance and predictive accuracy.

- Compressive strength (MPa): Indicates the load-bearing capacity and overall stiffness of printed layers.
- Flexural strength (MPa): Represents resistance to bending and interlayer failure [1].
- Tensile strength (MPa): Measures the interlayer bonding quality and crack resistance [2].
- Layer deformation (mm): Quantifies vertical settlement and layer displacement, used for real-time stability monitoring.
- Shrinkage strain (%): Captures both plastic and drying shrinkage responsible for dimensional instability [3].
- Crack width and propagation rate (mm/s): Measure structural integrity and durability of printed elements.

The integrated dataset will serve as the foundation for training and validating machine learning models capable of predicting deformation, shrinkage, and crack initiation in 3D-printed concrete, thereby supporting adaptive control mechanisms during the printing process.

D. Prediction Model Design

- **Data Preprocessing and Normalization:** Standardization and feature scaling of input variables will be performed to ensure consistent learning behavior.
- **Model Training:** Supervised learning algorithms such as *Random Forest Regressor*, *XGBoost*, and *Artificial Neural Networks (ANN)* will be trained on the collected dataset.
- **Model Evaluation and Optimization:** Model performance will be validated using metrics such as the coefficient of determination (R^2), Mean Absolute Error (MAE), and Root Mean Square Error (RMSE). Hyperparameter optimization will be performed using grid search and k-fold cross-validation to enhance generalization.

Model Integration for Real-Time Prediction: The trained model will be integrated into the 3D printing control loop, as illustrated in Fig. 3. Real-time sensor data from the printing process (e.g., flow rate, extrusion pressure, and layer deformation) will be continuously fed to the model. The model will output predicted deformation and shrinkage levels, which will guide *adaptive adjustments* in printing parameters (such as print speed and extrusion rate) to maintain layer stability and minimize cracking.

Expected Outcome: The proposed model aims to achieve the following outcomes:

- Improved accuracy in predicting mechanical strength, deformation, and shrinkage.
- Enhanced real-time control of 3D concrete printing through continuous feedback and adaptive parameter tuning.
- Development of a robust predictive framework applicable to diverse material mix designs and environmental conditions.

V. Conclusion & Future Work

This study presents an **AI-based intelligent framework** for improving the reliability and performance of **3D Concrete Printing (3DCP)** through real-time monitoring and adaptive control. The proposed system integrates **machine learning algorithms** such as **Random Forest, XGBoost, and Artificial Neural Networks (ANN)** to predict **layer deformation**, **shrinkage**, and **crack initiation** based on material composition, process parameters, and environmental conditions. By dynamically adjusting key printing variables like **extrusion rate** and **print speed**, the framework aims to ensure layer stability, minimize deformation, and enhance structural durability.

The **future work** will focus on **gathering the experimental dataset**, followed by **model training and optimization**. Subsequent stages will involve **developing AI models to analyse and evaluate local deformations during printing and dynamically adjust printing speed or extrusion rate to maintain structural stability**, and **predicting shrinkage behavior and the onset of crack formation** in printed structures to ensure long-term durability and interlayer bonding integrity.

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